
EXEMPLE cu HashSet, LinkedHashSet, TreeSet

Set - nu pune duplicatele (in orice fel de Set)

HashSet - nu se pastreaza ordinea

LinkedHashSet - se pastreaza ordinea

TreeSet - se pune sortat dupa un anumit criteriu

(pentru TreeSet trebuie sa avem "implements Comparable<Masina>"
si sa avem @Override la compareTo)

Masina.java:

```
package org.example;
```

```
public class Masina implements Comparable<Masina>{  
    private int id;  
    private int HP;  
    private String marca;  
  
    public Masina(int id, int HP, String marca) {  
        this.id = id;  
        this.HP = HP;  
        this.marca = marca;  
    }  
  
    @Override  
    public String toString() {  
        return "Masina " + id + " cu " + HP + " cai putere si marca " + marca;  
    }  
  
    /*  
        Masinile se vor compara in functie de Horse Power in TreeSet  
    */  
    @Override  
    public int compareTo(Masina o) {  
        return Integer.compare(this.HP, o.getHP());  
    }  
}
```

```
// GETTERI SI SETTERI PENTRU TOATE .... (i-am sters)
}
```

Main.java:

```
package org.example;

import java.util.*;

public class Main {

    static void main() {

        Masina m1 = new Masina(1, 270, "Bentley");

        Masina m2 = new Masina(2, 880, "X6 ");

        Masina m3 = new Masina(3, 200, "GLE Coupe");
```

```
        //HashSet - va pastra elementele fara dubluri, DAR NU IN ORDINEA
        // IN CARE S-AU PUS
```

```
        Set <Masina> cars_set = new HashSet<>();

        cars_set.add(m1);

        cars_set.add(m2);

        cars_set.add(m3);

        System.out.println("Masinile in HashSet:");

        for(Masina m : cars_set) {

            System.out.println(m);

        }
```

```
        /* se va afisa:
```

```
        Masinile in HashSet:
```

```
        Masina 2 cu 880 cai putere si marca X6
```

```
        Masina 1 cu 270 cai putere si marca Bentley
```

```
        Masina 3 cu 200 cai putere si marca GLE Coupe
```

```
        */
```

```
// HashSet - va pastra elementele fara dubluri, in ordine
```

```
Set <Masina> masinas = new HashSet<>();  
masinas.add(m1);  
masinas.add(m1);  
masinas.add(m1);  
masinas.add(m2);  
masinas.add(m3);  
System.out.println("Masinile in HashSet:");  
for(Masina m : masinas) {  
    System.out.println(m);  
}  
  
/* se va afisa  
Masinile in HashSet:
```

Masina 1 cu 270 cai putere si marca Bentley

Masina 2 cu 880 cai putere si marca X6

Masina 3 cu 200 cai putere si marca GLE Coupe

**/*

```
//TreeSet - va pune toate masinile (fara duplicate)
```

```
//pune in ordine crescatoare dupa Horse Power
```

```
Set <Masina> masinas2 = new TreeSet<>();  
masinas2.add(m1);  
masinas2.add(m2);  
masinas2.add(m3);  
masinas2.add(m3);  
System.out.println("Masinile in TreeSet:");  
for(Masina m : masinas2) {  
    System.out.println(m);  
}
```

/ se va afisa*

Masinile in TreeSet:

Masina 3 cu 200 cai putere si marca GLE Coupe

Masina 1 cu 270 cai putere si marca Bentley

Masina 2 cu 880 cai putere si marca X6

**/*

}

}